



ROOT CAUSE UNLOCKED

# Unlock Your *Mast Cell* Response

*A root cause roadmap for mast cell activation and histamine intolerance: what is actually being released, which tests separate the patterns, and how to lower the load*

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · [rootcauseunlocked.com](http://rootcauseunlocked.com) · (919) 662-0520

# Unlock Your Mast Cell Response

*A root cause roadmap for mast cell activation and histamine intolerance: what is actually being released, which tests separate the patterns, and how to lower the load*

## BEFORE YOU START

# Contents

Important Medical and Legal Disclaimer

---

How to Use This Book

---

Contents

---

## INTRODUCTION

Why the same protocol fails half the people who try it

---

1. What mast cell activation actually is, and the four things it gets confused with

---

2. What a mast cell releases, in three waves

---

3. The triggers that have nothing to do with allergy

---

4. Building a sound evaluation

---

5. The two exits: how your body clears histamine

---

6. Lowering the load: the diet levers

---

7. Raising clearance capacity

---

8. Calming the cell: what is worth trying and what is not

---

9. The medication conversation, and whose job it is

---

10. Safety: the reaction plan, the bleeding question, and when to escalate

---

11. The downstream branch: heavy periods, low iron, B12, and sleep

---

12. A ninety day roadmap

---

13. Worksheets

---

14. Closing perspective

---

Working with Root Health

---

References

---

# Important Medical and Legal Disclaimer

---

This book teaches the approach I use with patients whose symptoms look like mast cell activation or histamine intolerance. It is intended for education and general wellness support. It does not diagnose mast cell activation syndrome, hereditary alpha tryptasemia, systemic mastocytosis, histamine intolerance, allergy, or any other medical condition. It is not a substitute for examination, diagnosis, or treatment by a qualified healthcare professional, and reading it does not make you my patient.

Symptoms attributed to mast cells and histamine also occur with true IgE allergy, asthma, carcinoid and other neuroendocrine tumors, pheochromocytoma, thyroid disease, medication reactions, hereditary angioedema, autonomic disorders, anemia, infection, panic, and occupational exposure. New, severe, rapidly worsening, or unexplained symptoms should be evaluated promptly. Ruling these out is not a delay tactic. It is the foundation everything else in this book is built on.

**Anaphylaxis is a medical emergency.** Difficulty breathing, throat or tongue swelling, a drop in blood pressure, fainting, or a reaction that keeps escalating after the trigger is removed means calling 911, not opening a book. If you have ever had a reaction like that, the most important thing in this entire program is that you have an emergency plan and an epinephrine prescription from a physician. Nothing in these pages replaces either one.

Do not stop prescribed medication, begin high-dose supplements, undertake prolonged fasting or elimination, or start aggressive detoxification programs without professional guidance. Pregnancy, breastfeeding, kidney disease, liver disease, cardiovascular disease, bleeding disorders, mast cell disease under specialist care, and medication use can materially change the safety of anything discussed here.

# How to Use This Book

---

Read Chapters 1 through 3 before you change anything. They exist to keep you from spending a year and a great deal of money treating the wrong mechanism, which is the most common thing that happens to people in this situation.

Then read Chapter 4 and stop. That chapter is the evaluation, and the evaluation comes before the plan. Almost everything that goes wrong in this space goes wrong because someone started a twenty-agent protocol before anyone knew which mediator was actually driving their symptoms.

Chapters 5 through 9 are the plan itself, and they are ordered deliberately. Chapter 10 is the safety layer, which is not optional and is not last because it matters least. Chapter 11 covers the downstream consequences, the ones that show up as low iron and exhaustion rather than as flushing. Chapter 12 is a ninety day roadmap and Chapter 13 is the worksheets you will actually use.

If you only have twenty minutes, read the section in Chapter 4 called “Antihistamines that did nothing are telling you something.” It is the single highest-value page in this book, it costs nothing, and it redirects more people than any test I order.

# Contents

---

1. What mast cell activation actually is, and the four things it gets confused with
2. What a mast cell releases, in three waves
3. The triggers that have nothing to do with allergy
4. Building a sound evaluation
5. The two exits: how your body clears histamine
6. Lowering the load: the diet levers
7. Raising clearance capacity
8. Calming the cell: what is worth trying and what is not
9. The medication conversation, and whose job it is
10. Safety: the reaction plan, the bleeding question, and when to escalate
11. The downstream branch: heavy periods, low iron, B12, and sleep
12. A ninety day roadmap
13. Worksheets
14. Closing perspective

# Why the same protocol fails half the people who try it

---

Most people arrive at this topic the same way. Something started reacting. Foods that were fine became not fine. Wine, aged cheese, leftovers, a hot shower, a stressful week, a new supplement, a scent in an elevator. Somewhere in the search a word appeared, usually MCAS or histamine intolerance, and it explained enough of the picture to feel like an answer.

Then came the protocol. Almost always the same one: an antihistamine, quercetin, vitamin C, a low histamine food list, DAO capsules, and a long list of things to avoid. Some people improve dramatically on that. Others improve slightly. And a large group improves not at all, concludes they must be an especially severe case, and doubles the same protocol.

Here is what I think is actually happening in that third group, and it is the reason this book is organized the way it is.

**Not every mast cell symptom is a histamine symptom.** When a mast cell activates it does not release one thing. It releases a first wave of preformed contents in seconds, a second wave of newly built lipid mediators over minutes, and a third wave of cytokines over hours. Histamine is in the first wave only. Prostaglandin D2 and the cysteinyl leukotrienes are in the second, and not one of them is touched by an antihistamine. If your dominant mediator is prostaglandin D2, you can take antihistamines forever and stay exactly where you are.

**And not every histamine symptom is a mast cell symptom.** Histamine you eat is pharmacologically real. Seventy-five milligrams of oral histamine, a dose you can reach in one ordinary high histamine meal, produced immediate or delayed symptoms in five of ten healthy women with no history of intolerance at all (PMID 15603203). That is not a disease. That is a dose. Which means a person can have genuine histamine symptoms with completely normal mast cells, and the entire fix lives in the kitchen and in the enzyme that clears what arrives.

So there are at least two different problems wearing the same set of symptoms, and the interventions that fix one do nothing for the other. Sorting out which one you have is not an academic exercise. It is the difference between a plan that works and a plan that consumes two years.

That sorting is what I do first, and it is what this book teaches you to do with me.

# 1. What mast cell activation actually is, and the four things it gets confused with

---

## The definition that is actually used

Mast cell activation syndrome has a formal definition, and it has three parts, all of which have to be present (PMID 31476322, PMID 31470118). First, recurrent symptoms in two or more organ systems consistent with mast cell mediator release. Second, an increase in a mast cell mediator, measured during or shortly after an event, above that person's own baseline. Third, a response to therapy directed at mast cell mediators.

Read that second criterion again, because it is the one that gets skipped. The standard is a **rise above your own baseline**, conventionally 20 percent plus 2 ng/mL within about four hours of an event, not a single number crossing a population cutoff (PMID 34166845). A tryptase of 14 drawn on a random Tuesday proves nothing on its own. A tryptase that goes from 6 to 9.2 during a reaction is a different piece of information entirely.

I am telling you this early, in plain language, because I have met a lot of people who were handed a diagnosis on a single elevated number and a lot of others who were told they could not possibly have a mediator problem because one number was normal. Both of those conclusions are built on the same misunderstanding.

## Confusion one: hereditary alpha tryptasemia

Roughly four to six percent of the general population carries extra copies of the alpha tryptase gene at TPSAB1. Those people run a persistently elevated baseline tryptase, commonly in the 8 to 15 ng/mL range, with no clonal mast cell disease and, often, no mast cell activation at all (PMID 27749843).

This is not rare. In a room of a hundred people, four or five of them have it. It is the single most common reason a tryptase comes back high, and if nobody checks for it, a perfectly ordinary tryptase elevation becomes a diagnosis, and sometimes becomes a bone marrow biopsy. Using genotype-defined individual reference ranges has been shown to prevent unnecessary procedures (PMID 36170795). It also matters in people who genuinely do have mast cell disease, where extra alpha tryptase copies independently correlate with disease features (PMID 36309122).

If your tryptase is elevated and nobody has raised this with you, that is the next conversation to have, before anything else.

## Confusion two: histamine intolerance

Histamine intolerance is a clearance problem, not an activation problem. The mast cells are behaving. The issue is the volume of histamine arriving from food and from gut bacteria against the capacity of the enzyme

that degrades it.

The reason this matters so much is that it responds to completely different levers. A histamine intolerance picture responds to lowering dietary amine load, to repairing the gut lining, and sometimes to a swallowed enzyme. It does not need a mast cell stabilizer, because nothing is destabilized.

### **Confusion three: allergy**

Allergy is IgE against a specific antigen, reproducible, testable, and usually rapid. Mast cell activation of the kind this book is about frequently involves no IgE whatsoever, which is why allergy testing so often comes back clean and the person is told nothing is wrong. Chapter 3 covers the non-IgE routes in detail, because they are the ones that explain the reactions that make no sense.

### **Confusion four: systemic mastocytosis**

Mastocytosis is a clonal disease. There are too many mast cells because of a driver mutation, most often KIT D816V. It is diagnosed by bone marrow evaluation and it is managed by hematology. It is not the same thing as mast cells that are normal in number and overly reactive. A very high baseline tryptase, unexplained bone lesions, unexplained low blood counts, an enlarged liver or spleen, or the characteristic skin lesions are all reasons to be evaluated for it rather than managed with diet.

### **Why I make you sit through this**

Because the plan is different for each one. And because the honest version of this field, which almost nobody says out loud, is that the label matters less than the mechanism. I do not particularly need to win an argument about whether you meet consensus criteria. I need to know which mediator is driving your symptoms and where the load is coming from. That is answerable.

## 2. What a mast cell releases, in three waves

---

A mast cell is a long-lived tissue resident cell that sits exactly where the outside world meets you: skin, airway, gut lining, and around blood vessels. It is not a circulating cell you can count in a blood draw. When it activates, it releases in three waves.

**Wave one, seconds.** The preformed granule contents. Histamine, tryptase, chymase, carboxypeptidase, and heparin, all packaged together on a scaffold. Mast cells are also unusual in storing tumor necrosis factor ready to go.

**Wave two, minutes.** Newly built lipid mediators. Arachidonic acid is liberated from the membrane and converted down two branches. One branch makes prostaglandin D2 (PMID 12895599). The other makes the cysteinyl leukotrienes, LTC4 to LTD4 to LTE4. Platelet activating factor is made here too.

**Wave three, hours.** Cytokines and chemokines, including interleukin 6, interleukin 8, and TNF.

### The part that changes what you do

Antihistamines block histamine receptors. That is all they do. They do not touch prostaglandin D2, they do not touch leukotrienes, and they do not touch platelet activating factor, which is itself implicated in antihistamine-refractory urticaria (PMID 39596211).

Prostaglandin D2 acts on two receptors and produces the flush, vasodilation, and low blood pressure picture. Cysteinyl leukotrienes act on their own receptors and produce bronchoconstriction, vascular leak, mucus, and a cramping gut-dominant symptom picture. Prostaglandin D2 also contracts uterine smooth muscle, which is a large part of why some women cluster symptoms around the luteal phase and menses and cannot work out why their protocol stops working for a week every month.

So when someone tells me antihistamines did nothing, I do not conclude they were wrong about mast cells. I conclude their dominant mediator is probably not histamine, and the next step is not a third antihistamine.

### Release is not all or nothing

There are two modes. Full degranulation, where granule membranes fuse and everything empties in seconds, which is what happens in anaphylaxis. And piecemeal or selective release, which is slow, partial, differential, and leaves no visible granule loss.

This is the mechanism behind one of the most demoralizing experiences in this condition: chronic daily symptoms with a completely normal tryptase. A normal tryptase does not mean nothing is being released. It means the release mode was not the kind that dumps tryptase into your blood.

## **Tryptase, heparin, and one thing worth knowing**

Baseline tryptase indexes how many mast cells you have in total. It is a useful number, with the alpha tryptasemia caveat from Chapter 1 attached to it permanently.

Heparin is the one people miss. Mast cell granules are the only significant store of true heparin in the human body, and it is biologically active the moment it is released. Clotting abnormalities are documented in mastocytosis (PMID 26447996), heparin-like anticoagulant activity has been caught on thromboelastography during anaphylaxis-associated coagulopathy (PMID 39557214), and tryptase itself alters clotting kinetics (PMID 12701115). A review of primary mast cell disease documents easy bruising and excessive bleeding as frequent findings (PMID 34445665).

Practically: if you bruise easily, bleed longer than you think you should, or are heading into surgery or a dental procedure, that is mediator information and your surgeon should know about the mast cell picture. It is not incidental.

### 3. The triggers that have nothing to do with allergy

---

This is the chapter that explains the reactions that make no sense, and it is the reason so many people are told their testing is normal and therefore nothing is happening.

There are at least five ways to activate a mast cell, and only the first one is allergy.

**One: IgE crosslinking.** The classic allergic route. Real, testable, and not what is going on in most of the people reading this.

**Two: MRGPRX2.** A receptor found specifically on mast cells that responds to a whole class of positively charged compounds with no IgE involved at all: opioids, fluoroquinolone antibiotics, vancomycin, neuromuscular blockers used in anesthesia, and the body's own neuropeptide substance P (PMID 25517090). This is the molecular explanation for "pseudo-allergic" drug reactions, and it is why a reaction to a medication can be entirely real and entirely invisible on allergy testing.

**Three: osmolality.** This one is badly under-appreciated and it is directly relevant to anyone taking supplements. Concentrated solutions trigger mast cells directly and in a dose-dependent way. Nasal instillation of a hyperosmolar mannitol solution in fifteen healthy volunteers released histamine and leukotrienes compared with an isosmolar control, with more osmolality producing more histamine (PMID 2449834). In isolated human bronchi, the same stimulus produces mast cell dependent contraction (PMID 31207273). This is the accepted mechanism behind exercise-induced bronchoconstriction.

Now think about what that means for a protocol. Concentrated powders taken in a small amount of water, undiluted liquid supplements, hypertonic nasal rinses, and osmotic laxatives all raise local osmolality. I have watched people conclude they react to an ingredient when what they were reacting to was the concentration. Dilution is free, and it is the first thing I change.

**Four: physical inputs.** Mechanical pressure, vibration, heat, and cold get converted directly into the same activation signal. A single genetic variant in a mechanosensing receptor is enough to cause human vibratory urticaria all by itself (PMID 26841242).

**Five: the nervous system.** Corticotropin releasing hormone, the front end of the stress axis, degranulates human nasal mucosal mast cells in laboratory conditions (PMID 33803422) and rodent skin mast cells in a living animal with measurable increases in vascular permeability (PMID 9421440). Substance P works through the MRGPRX2 receptor above.

I want to be precise about why this fifth one matters, because people in this situation have usually been told their symptoms are stress, meaning imaginary. That is not what this says. This says stress hormones physically degranulate mast cells through a receptor, bypassing IgE entirely. The symptom is not in your head. The trigger has a molecular address. That is a completely different statement, and it means nervous system work is not a consolation prize. It is acting on a documented input.

## **The brakes your body already has**

The system has built-in brakes, which is useful to know because two of them are things you can influence.

Prostaglandin E2 prevents hyperosmolar activation of human mast cells (PMID 25329458), and blocking that pathway with an anti-inflammatory makes the mannitol response worse rather than better (PMID 31207273). The vitamin D receptor physically blocks a key signaling step in mast cell activation, and mast cells spontaneously activate in a vitamin D deficient environment (PMID 27998003). And a family of locally released fat-derived compounds, of which palmitoylethanolamide is the best known, act as local brakes. Impaired release of that compound in the duodenum is what underlies acid-induced mast cell activation in human functional dyspepsia (PMID 33065341).

Two of those three are actionable, and they come back in Chapter 8.

## 4. Building a sound evaluation

---

### **Antihistamines that did nothing are telling you something**

Start here. If you have taken a standard second generation antihistamine consistently, at an adequate dose, for long enough to judge it, and it did nothing at all, you have generated real diagnostic information for free: your dominant mediator is probably not histamine.

That single observation redirects the plan. It moves attention toward the prostaglandin and leukotriene arms, which need a different test and a different class of intervention entirely, and it tells us that stacking more histamine-directed agents is unlikely to be the answer.

### **The baseline tryptase, and why it must be drawn when you are well**

A baseline tryptase drawn on a symptom-free day is the reference point that makes every future value interpretable. Without it, an acute value has nothing to be compared against, and the criterion that actually matters, a rise of 20 percent plus 2 ng/mL over your own baseline within about four hours, cannot be applied (PMID 34166845).

If it comes back elevated, the very next question is hereditary alpha tryptasemia, per Chapter 1. Not mastocytosis, not MCAS. Alpha tryptasemia first, because it is far more common and because the test for it exists.

### **The twenty-four hour urine mediator panel**

This is the test that separates the phenotypes, and in my view it is the single highest-yield thing you can order in this space.

It measures three arms at once (PMID 35346887). The prostaglandin D2 metabolites, 11-beta-PGF2-alpha and its dinor metabolite. LTE4, the stable urinary form of the cysteinyl leukotrienes. And N-methylhistamine, the validated readout of histamine turnover (PMID 8859283, PMID 39745330).

Running all three tells you which arm is actually driving your symptoms, and therefore which class of intervention has any chance of working. A patient whose leukotriene metabolite is high and whose histamine metabolite is normal is being failed by a histamine protocol, and no amount of persistence will change that. A patient whose prostaglandin arm is dominant needs a completely different conversation. Current reviews of mast cell biomarkers reinforce that these urinary measures carry information tryptase alone does not (PMID 39745330).

I am aware this test is not universally available and not universally cheap, and I am not going to pretend otherwise. Where it cannot be obtained, we make decisions from the response pattern instead, which is less precise and still far better than treating all three phenotypes identically.

## Plasma histamine, and why I rarely order it

Plasma histamine did not move at all in healthy volunteers after a 75 mg oral histamine dose that produced actual symptoms (PMID 15603203). A test that does not move when the thing it measures is demonstrably doing something is not a useful test for that purpose. Whole blood histamine has a separate problem: it tracks how many basophils you have, at roughly 1 to 2 picograms per basophil, and it falls when basophil counts fall for reasons that have nothing to do with your histamine handling (PMID 2477430).

## Serum DAO, read as a stratifier and never as a diagnosis

Diamine oxidase is the enzyme that clears histamine outside your cells, and there is a blood test for it. It is worth understanding what that number can and cannot do.

In a series where patients with suspected histamine intolerance underwent a placebo-controlled challenge, serum DAO tended lower in the confirmed cases but only at  $p$  equal to 0.08, and was highly variable in the people who did not have the condition (PMID 37648152). It is not established that serum DAO reflects gut DAO (PMID 33921522), and it did not reliably move on a low histamine diet in patients who had the diagnosis (PMID 38769188).

Two confounders hit hard and both are common. Heparin releases DAO that is stored bound to the vessel wall, so anyone who has recently had heparin reads spuriously high (PMID 32985651, PMID 2110916). And the placenta expresses enormous amounts of it, so plasma DAO in pregnancy is uninterpretable for this purpose (PMID 18499706).

I use it as one input among several, most usefully to decide whether a trial of the oral enzyme is worth running, because that is where the benefit consistently concentrates. I do not diagnose anyone with a number from this test.

## Genetics, honestly

AOC1 variants that lower DAO activity are common. Over seventy percent of people of European ancestry carry at least one, and one specific allele lowers serum DAO with a clear gene-dose effect (PMID 17700358). A study of 484 people found seven variants strongly associated with DAO activity, and here is the part that matters: those variants **were not associated with the histamine intolerance phenotype**, only with lower enzyme activity (PMID 21488903).

Genotype tells you who has less enzyme. It does not tell you who is ill. I treat it as a stratifier that helps me set expectations, never as a diagnosis.

The same discipline applies on the intracellular side. A common variant in HNMT, the other clearance enzyme, lowers activity by raising the enzyme's requirement for its methyl donor (PMID 9547362, PMID 11566133), and complete loss of that gene causes a serious neurological condition, which is the cleanest available proof that the enzyme genuinely matters in the human brain (PMID 26206890). Interesting, real, and still not a diagnosis by itself.

## The differential that outranks the histamine story

Before we commit to this model, some other things need to be off the table, because they are more common, more treatable, and more consequential if missed.

If iron deficiency is part of your picture, celiac disease, autoimmune atrophic gastritis and *Helicobacter pylori* are the classic occult causes and they need to be excluded (PMID 15921373, PMID 24215034).

Autoimmune atrophic gastritis is worth naming twice, because it is the one diagnosis that explains both the iron arm and the B12 arm at once.

If heavy menstrual bleeding is part of your picture, structural causes and von Willebrand disease outrank mast cells by a wide margin. Von Willebrand disease is a leading inherited cause of heavy menstrual bleeding, and women who have it carry high rates of heavy bleeding with iron deficiency (PMID 40697484, PMID 40955497).

If unrefreshing sleep, morning headache, and daytime exhaustion are part of your picture, obstructive sleep apnea needs screening, for reasons covered in Chapter 11.

None of this is me talking you out of a mast cell problem. Plenty of people have both. It is me refusing to let a treatable diagnosis hide behind an untreatable-sounding one.

## For the chemically reactive reader

If your dominant story is reacting to fragrances, cleaning products, and building materials rather than to foods and physical triggers, I want to be straight with you about the evidence, because a great deal of the internet is not.

There is no direct human evidence that mast cells mediate multiple chemical sensitivity. Double-blind placebo-controlled chamber provocation in patients with that presentation does not reproduce symptoms under masked exposure (PMID 18568800), and a systematic review of thirty-seven provocation studies covering 784 patients found that reactions track whether the person can tell the active substance from the sham (PMID 17137865).

Only one published study has measured histamine in that population, and what it found is the important part: substance P, vasoactive intestinal peptide and nerve growth factor were elevated, **and histamine was not**. Solvent exposure then raised all of them in those patients but not in controls, and increased how reactive the skin was to histamine applied directly (PMID 15031958). Substance P activates mast cells, so there is a genuine route from an irritated sensory nerve to a degranulating mast cell. But it runs nerve first and mast cell second. In that presentation histamine looks like a passenger, sometimes an amplifier, not the engine.

What does have blinded, objective support there is sensory nerve sensitization. Patients meeting chemical sensitivity criteria coughed significantly more than controls at every dose of inhaled capsaicin in a randomized double-blind trial, below the concentration anyone can taste or smell (PMID 12448352), and showed a significantly larger area of spreading sensitivity on quantitative sensory testing (PMID 22487274).

So this is not a dead end, it is a different door. If fragrances and cleaning products are your story rather than foods and physical triggers, the work that pays is calming the nerve rather than blocking the histamine, and chemical sensitivity has its own approach and its own guides built around exactly that. I would rather point you there than sell you a mast cell certainty I do not have.

# 5. The two exits: how your body clears histamine

---

Histamine leaves by two routes, and they are not backups for each other. They serve different compartments (PMID 17490952).

**The outside route is diamine oxidase.** DAO is a secreted enzyme concentrated in the tips of the villi in your small intestine, plus kidney and placenta. It handles histamine that is eaten and histamine released into the space between cells. It is a copper enzyme, and its catalytic machinery is built in place in a reaction that requires copper twice over (PMID 20722416, PMID 27063138).

That detail matters commercially, because DAO is **not** a vitamin B6 enzyme, and a great deal of consumer marketing says otherwise. The B6 dependent enzyme in this system is histidine decarboxylase, the enzyme that **makes** histamine (PMID 413882, PMID 7075603). Getting that backwards is not a small error.

**The inside route is HNMT.** Histamine that is inside a cell is instead ring-methylated by histamine N-methyltransferase, a cytosolic enzyme (PMID 7943261). Because it is inside cells and because the brain has essentially no DAO, the entire central nervous system depends on this route, and it depends on transporters to get histamine into the cell in the first place (PMID 27679412, PMID 33641029). So does the airway: blocking HNMT potentiates histamine-induced contraction of isolated human bronchi, while blocking DAO does not (PMID 7943261).

If your symptoms are cognitive, neurological, or airway-dominant, the intracellular route is the one governing them, and diet plus a swallowed enzyme is acting on the other compartment.

## The single most useful thing to understand about DAO

DAO capacity is essentially a function of healthy gut lining. The enzyme lives in the villous tips, so the amount of enzyme you have tracks how much intact absorptive surface you have.

The evidence for that is unusually clean. Post-heparin plasma DAO correlates with jejunal mucosal DAO at  $r$  equal to 0.84. It is markedly reduced in active Crohn's disease in proportion to how active the disease is, and returns toward normal in remission. And it drops dramatically during total parenteral nutrition and reverses when oral feeding resumes (PMID 2110916). In celiac disease, post-heparin DAO rose into the normal range within three months of a gluten free diet as the villi recovered (PMID 3121453), and reduced DAO shows up in both serum and biopsy tissue in childhood celiac disease (PMID 3088248). The Crohn's series did not reproduce a celiac deficit in its own subgroup, so the celiac literature is not unanimous, and I will say that rather than overstate it.

## And the part that saves you money

This node behaves like a floor, not a dial. Copper deficiency lowers DAO. Copper **above** adequacy does not raise it. In a controlled animal study, a frankly copper deficient diet dropped DAO, while supplementing

copper at roughly five times the adequate intake changed nothing at all (PMID 10953681).

So there are exactly four ways to move this system, and only four. Repair the mucosa. Correct a genuine copper deficiency. Put exogenous enzyme in the gut lumen. Or reduce the substrate arriving. Everything else being sold for “DAO support” is acting on a system that does not respond that way.

## **The methylation exit, and the honest limits of the methylation conversation**

HNMT transfers a methyl group from SAM, and the reaction leaves behind SAH. Human kinetics show SAM binds first and SAH leaves last, and critically, **SAH competitively inhibits the enzyme** (PMID 7188427). So the variable that governs this route is the SAM to SAH ratio, not homocysteine on its own.

That same study found something worth sitting with: histamine itself inhibits the enzyme at high concentrations. During a high histamine flare, the intracellular clearance route gets worse, not better. That is a real, measured reason why flares compound.

Where the methylation conversation goes wrong is in the leap from there to a personality-type framework. The “overmethylator” and “undermethylator” constructs have no published definition, no validated test, and no biomarker. Whole blood histamine cannot serve as one, for the basophil reason above (PMID 2477430). Low homocysteine has no validated reference interval that assigns it that meaning. The only validated methylation index is the plasma SAM to SAH ratio, and a 2025 randomized trial lowering SAH by about 12 percent and raising the ratio by about 26 percent is the first demonstration that the index can even be moved with nutrition (PMID 40883125).

What is defensible is narrower and more useful. Riboflavin at 1.6 mg per day raised SAM specifically in people with the MTHFR 677TT genotype in a randomized trial (PMID 32330571), which is the cleanest human demonstration that a B vitamin can move the variable this enzyme actually depends on. Homocysteine’s exits require B12 and folate on one route, betaine on another, and B6 on the transsulfuration route (PMID 9398304, PMID 9681996). And HNMT is competing for methyl groups against much larger consumers, including creatine synthesis, phospholipid methylation, and nicotinamide clearance (PMID 9597750, PMID 26598833, PMID 28291578, PMID 24717514). That competition is real. It is also not something you can currently measure in a way that would let anyone promise you a result.

## 6. Lowering the load: the diet levers

---

The full detail lives in the diet guide. This chapter is the reasoning, because the reasoning is what lets you make decisions when a food list disagrees with itself.

### Start with the finding that should humble everybody

In fifty-nine patients with suspected histamine intolerance, a single-blind placebo-controlled oral histamine challenge **excluded** the diagnosis in fifty of them, roughly 85 percent. Thirty-seven of the fifty-nine, about 63 percent, reacted to the placebo. Only about 7 percent had objective symptoms to histamine and not to placebo (PMID 37648152).

And three quarters of those patients had previously reported improving on a low histamine diet.

I am not showing you this to talk you out of trying the diet. I am showing it to you because it establishes something that will save you enormous grief: **feeling better on a low histamine diet is not proof that histamine was the mechanism**. Expectation is powerful here, and the trial design that separates the two is not one you can run at home. So run the diet, use it, and hold the conclusion loosely.

### Lever one: preformed histamine in the food

Food is a bioreactor. Fresh protein carries histidine mostly bound up in protein and makes a poor histamine source. Ripening, aging, fermentation, and bacterial breakdown liberate free histidine, and then bacterial enzymes decarboxylate it into histamine. Those bacterial enzymes come in two unrelated families, which is why no single nutrient switches them off (PMID 18756395, PMID 2857718, PMID 3082865).

Two properties make this practically important. Histamine is heat stable, so cooking kills the organism and leaves the product intact (PMID 25876709). And refrigeration does not stop it: cold-tolerant bacteria keep producing histamine at 1 to 3 degrees C, and vacuum-packed tuna held near 2 degrees C reached over 5,000 mg/kg (PMID 15925710, PMID 18845350).

Which is why storage rules and leftover rules do more for some people than any supplement in this book, and cost nothing.

### Lever two: total amine load, not histamine alone

DAO is saturable and it is not selective. Putrescine, cadaverine and tyramine in the same meal compete with histamine for the same active site and slow its degradation (PMID 35694170, PMID 40865824).

This resolves a puzzle that frustrates a lot of people. Across ten published low histamine diets, only about 32 percent of the excluded foods were excluded because of their histamine content (PMID 33919293). Bananas and citrus appear on the lists despite containing very little histamine. They are not on there for their

histamine. They are on there for competition and for other reasons, and understanding that lets you reason about a meal rather than obey a list.

### **Lever three: your own gut bacteria**

Some of the histamine is being made downstream of your mouth. Fecal bacteria from IBS patients with high urinary histamine produce large amounts of histamine in the laboratory, and a specific organism carrying a histamine-producing gene variant was identified as a major producer and was over-represented in three independent human patient groups (PMID 35895832). Case-control work in histamine intolerance finds a consistent pattern: more of certain histamine-producing organisms, fewer of the protective ones (PMID 35565742, PMID 30552302).

This gives you a lever that no food list gives you. In a randomized trial, three weeks of a low fermentable carbohydrate diet versus a high one reduced urinary histamine roughly eightfold (PMID 26976734). That is the only randomized trial anywhere in this literature that used a histamine biomarker as an actual outcome, and it works by starving the bacterial enzyme of substrate rather than by removing preformed amine from your plate. It is a genuinely different lever, and I use it in people whose picture is gut-dominant.

### **The fermented food conflict, stated rather than resolved**

All ten published low histamine diets exclude fermented foods, on measured amine content (PMID 33919293). Against that, a seventeen week randomized trial of six servings of fermented foods per day raised microbiota diversity and lowered nineteen inflammatory proteins in healthy adults, without measuring histamine at all (PMID 34256014).

Those two findings point in opposite directions and I am not going to pretend one of them does not exist. Fermented food is probably good for a general population and probably a poor idea during an active elimination phase in an amine-loaded patient. This matters practically because gut repair programs quietly push fermented food, which can undo the diet you are running. Reintroduction, later and deliberately, is where I put it.

### **Alcohol, counselled correctly**

Alcohol belongs in this conversation, but for the right reasons. The claim that ethanol and acetaldehyde inhibit DAO does not survive testing: neither, nor disulfiram, inhibited recombinant human DAO at physiologically relevant concentrations, and the apparent disulfiram effect turned out to be an assay artifact (PMID 27401969). Alcohol earns its place on amine content and on mast cell provocation, not on DAO inhibition.

## And the drug list, mostly folklore

The circulating “drugs that block DAO” lists are largely unsupported. Tested directly, ibuprofen, aspirin, paracetamol, sumatriptan and zolmitriptan did not inhibit DAO. Only naproxen lowered DAO protein, and it did so without affecting activity (PMID 38068554). The node is genuinely blockable by some compounds (PMID 3134804, PMID 2568741, PMID 2128501), but the list being passed around is not that list.

There is one drug screen that is not folklore and is a safety matter rather than a comfort matter: monoamine oxidase inhibitors and tyramine. Phenelzine, tranylcypromine, isocarboxazid, high-dose selegiline and linezolid combined with a tyramine-rich meal can produce a hypertensive crisis (PMID 3525654, PMID 17192761). Tyramine shares a food list with histamine and shares nothing else mechanistically. If you take any of those, that is a conversation with your prescriber before you touch aged foods.

## One more source people forget

The stomach has its own histamine story, and it is mostly about acid secretion rather than about your systemic burden. *Helicobacter pylori* activates the histidine decarboxylase promoter in gastric cells (PMID 10652359), but human oxyntic mucosal histamine content was actually **lower** in infected duodenal ulcer patients and rose after eradication (PMID 8338090), and gastric histamine concentrations in infection are not uniformly elevated (PMID 1856297). I mention it because it gets cited both ways online, and the honest reading is that gastric histamine physiology is a different question from the one this book is about.

## 7. Raising clearance capacity

---

Four levers, in the order I use them.

### **Repair the lining**

Given that DAO capacity tracks intact villous surface (PMID 2110916), anything that damages the small intestinal lining lowers your ceiling. Which is why celiac screening is on the evaluation list rather than being optional, why active inflammatory bowel disease has to be managed by the people who manage it, and why gut work is not a side quest here. It is the main lever on this enzyme.

### **Do not starve the enzyme by starving yourself**

Post-heparin DAO was 4.5 during total parenteral nutrition against 35.9 in controls, and it reversed on resumption of oral feeding (PMID 2110916). Food in the lumen maintains the cell mass that **is** the enzyme.

I state this as a caution and not as demonstrated harm, because intravenous feeding is a far more extreme insult than an elimination diet. But the direction of the finding should worry anyone who has been on a narrow diet for a year. A prolonged, very restricted diet may be eroding the clearance capacity being blamed for the problem. Every restriction in my plans has an end date and a breadth floor for this reason.

### **Copper at adequacy, and no more**

Copper is required for the enzyme. It is a floor, not a dial (PMID 10953681). Correct a real deficiency, do not chase a benefit above adequacy, and watch the zinc side of the ledger, because sustained high-dose zinc induces copper deficiency. That is the actual copper problem I see most often, and it comes from a zinc supplement.

### **Oral DAO enzyme, in the right person**

Swallowed diamine oxidase acts in the gut lumen on histamine arriving from food. The trial evidence is real and modest. Two randomized double-blind placebo-controlled trials in adjacent populations (PMID 29698966, PMID 29475774), a genotype-selected trial in insomnia symptoms (PMID 41437955), and an open-label withdrawal study (PMID 31807350).

The honest limits: all small, all short, most conducted close to industry, and in the migraine trial the between-group difference did not reach significance. The consistent signal across them is that benefit concentrates in people selected for low baseline DAO. So this is a targeted trial in a selected person, not a default addition, and I run it as a trial with a defined endpoint rather than as a permanent purchase.

## 8. Calming the cell: what is worth trying and what is not

---

The supplement guide has the doses and the sequencing. This chapter is the honest tiering, because knowing which of these actually has human evidence behind it is the thing that lets you spend your money in the right order.

### The evidence tier, stated plainly

Almost every trial I am about to cite was conducted in systemic mastocytosis, chronic spontaneous urticaria, allergic rhinitis, migraine, IBS, histamine intolerance, or healthy volunteers. Very little of it was conducted in a population defined the way you were probably defined. What you are getting from me is established mechanism, trials in adjacent populations, and reasoned clinical practice held to a measurable endpoint. It is not a set of outcome trials in your exact diagnosis, because those do not exist, and I am not going to imply otherwise.

### Where the human evidence actually is

**Palmitoylethanolamide.** This one surprises people, because it is not the famous one. In a 108-person double-blind randomized trial, 350 mg per day for fourteen days significantly lowered histamine, interleukin 4, interleukin 8, interleukin 10 and TNF-alpha compared with placebo (PMID 38068797). The caveat travels with it permanently: the primary symptom endpoint was negative overall and positive only in the moderate-severity subgroup. Even with that caveat, this is better human evidence than most of what gets sold in this category. It also has a mechanistic home, since impaired release of this exact compound underlies acid-induced mast cell activation in human functional dyspepsia (PMID 33065341).

**Vitamin D.** Mechanism plus a trial. The vitamin D receptor physically interferes with mast cell activation signaling, and mast cells spontaneously activate in a vitamin D deficient environment (PMID 27998003). In chronic urticaria, vitamin D as an add-on to antihistamine has randomized support (PMID 26097642). This is why I test rather than guess, and why I hold a target, which is 40 to 80 ng/mL on 25-OH vitamin D.

**Quercetin, with the form corrected.** The two positive randomized trials used **bioavailable** forms at 200 mg per day (PMID 19295240, PMID 35776034). A larger dose of a poorly absorbed form can deliver less active compound than a smaller dose of a good one. Form matters more than dose here, which is the opposite of most people's instinct. Two honest limits: quercetin has no evidence at the DAO clearance node despite being sold for it, and it inhibits iron absorption, so it never shares a dose with iron.

**Butterbur, with a genuine split verdict.** A standardized, pyrrolizidine-alkaloid-free Petasites extract is the best-evidenced botanical for allergic rhinitis symptoms: equivalent to cetirizine in a randomized trial (PMID 11799030), dose-dependent in a multicentre placebo-controlled trial (PMID 15611396), and it attenuates the histamine-induced skin response in humans (PMID 14989395). And when the same extract was tested specifically against intestinal histamine handling, it changed neither DAO protein nor activity, and at high

concentrations increased histamine flux across the barrier (PMID 35149385). So: reasonable for upper airway symptoms, not a DAO agent, and only ever in a certified alkaloid-free preparation because of liver toxicity in unpurified forms.

**Boswellia.** A randomized placebo-controlled trial of a Boswellia-containing combination built on 5-lipoxygenase inhibition exists (PMID 29210124). Small, industry-run, and a two-herb combination rather than Boswellia alone, so I hold it at modest confidence. It is the only botanical with a trial aimed at the leukotriene arm, which is why it appears at all.

## Where the evidence is thinner than the marketing

**Luteolin** is animal and cell work only, with no human trial measuring any mast cell endpoint. I will use it riding alongside PEA, and I will label it as what it is.

**Omega-3** has an important null result that is rarely quoted. Four grams of EPA plus two grams of DHA daily for three weeks did not change bronchial hyperresponsiveness or urinary mast cell mediator excretion in asthmatics, despite confirmed adherence (PMID 25321659). I still use omega-3 fats in plenty of people. I do not file them under mast cell stabilization, because the one human trial that measured that endpoint was negative.

**Magnesium** stays for its many other jobs. There is no human dose studied for a mast cell endpoint, and I will not imply one.

**Zinc is not what people think it is at this node**, and this is worth reading twice. Zinc chelation **blocks** IgE-induced mast cell degranulation, and human mast cells actively release zinc when their receptors aggregate (PMID 16818790, PMID 23711853). Zinc is permissive **for** degranulation, not protective against it. Zinc has real uses. Piling it on as a mast cell stabilizer is not one of them, and at sustained high doses it costs you the copper your DAO enzyme requires.

**The glutathione block** (N-acetylcysteine, liposomal glutathione, selenium, alpha lipoic acid, milk thistle) has no human study of any of those agents against any mast cell or DAO endpoint. There is a mechanistic wrinkle worth knowing about, which is that the enzyme that makes prostaglandin D2 is glutathione dependent (PMID 12895599) and the enzyme that makes LTC4 conjugates glutathione directly. I am not going to build a claim on that in either direction. I use these agents where they belong, which is the chemical sensitivity and detox work, not here.

**Binders** have no evidence at any node in this system and carry a real cost, which is that they bind minerals and medication indiscriminately. If they are used at all, they go two hours away from everything.

## The excipient problem nobody warns you about

Given MRGPRX2 and given osmolality, the capsule matters. Polysorbate 80, polyethylene glycol, and certain other excipients have challenge-confirmed reactions behind them. Concentrated powders taken in too little water raise local osmolality directly (PMID 2449834, PMID 31207273).

Practically, this means that when someone “reacts to a supplement,” at least three explanations are on the table before the active ingredient: the excipient, the concentration, and the speed of introduction. Dilute it, change the form, slow down, and see whether the reaction survives.

## 9. The medication conversation, and whose job it is

---

I am a chiropractor. I do not prescribe, and I am not going to pretend the prescription tier of this field does not exist, because for some people it is the tier that works. Here is what exists, why it is on the list, and what to do with it.

**Second generation antihistamines.** The over-the-counter ones you can buy are a reasonable trial you can run yourself, and they are the highest-tier intervention on the histamine arm. In mastocytosis, a randomized double-blind placebo-controlled crossover trial of an H1 antihistamine at 20 mg daily in thirty patients improved itch, wheal and flare, flushing, tachycardia and headache (PMID 23734572). The specific agent in that trial is not available in the United States, but the class is, and updosing second generation antihistamines up to fourfold is standard practice in chronic urticaria and has systematic review support (PMID 33030434). Updosing is a conversation to have with the physician managing that part of your care, not something to freelance.

**H2 blockers.** Famotidine is the standard second agent, and combined H1 plus H2 blockade was the active comparator arm in the 1985 mastocytosis trial (PMID 3917606). I prefer famotidine over cimetidine because cimetidine interferes with liver drug metabolism far more.

There is a real cost attached to this class that almost nobody mentions, and it comes back in Chapter 11: long-term acid suppression is itself a documented mechanism for B12 and iron deficiency (PMID 24327038, PMID 27890768).

**Oral cromolyn** is a prescription mast cell stabilizer that some physicians use for gastrointestinal-dominant mast cell disease. It is not something I prescribe, and it is included here so you know it exists and can raise it with the physician managing that part of your care. It carries the only double-blind placebo-controlled trial in a mast cell mediator disorder, at 200 mg four times daily (PMID 2110198). It is worth understanding why it works where it works: it is under one percent absorbed orally, so it is effectively a topical gut drug, which is exactly what that trial found, benefit for gastrointestinal symptoms rather than systemic ones.

**Leukotriene receptor antagonists** are prescription agents that some physicians use as add-on therapy in antihistamine-refractory urticaria and in mast cell disease. They are not something I prescribe, and they are included here so you know they exist and can raise them with the physician managing that part of your care. They have systematic review and trial support as add-ons (PMID 24817895, PMID 19085267) and appear as a standard tier in mast cell treatment algorithms (PMID 30961835). This is the class that matters most if your urine panel shows a leukotriene-dominant picture.

**Ketotifen** is a prescription agent used in some countries and compounded in others, appearing in mast cell algorithms. It is not something I prescribe, it is included here so you know it exists, and I will say plainly that I did not find trial evidence for it at the standard I applied to everything else in this chapter.

**Aspirin, and the trap attached to it.** In a prostaglandin-dominant patient, aspirin is the intervention that has actually been tested. Doses studied in systemic mastocytosis ran from 81 mg twice daily to 500 mg twice daily, normalizing the urinary prostaglandin metabolite in twenty of twenty patients (PMID 19429499), with meta-analytic evidence from the adjacent field of aspirin-exacerbated respiratory disease (PMID 31518069).

Now the part that must never be separated from it. In eight of ten mastocytosis patients whose prostaglandin metabolite normalized on aspirin, the urinary leukotriene metabolite rose roughly thirteenfold (PMID 34029712). Blocking one branch pushes substrate down the other. Aspirin and leukotriene blockade are a pair at this node, not alternatives, and the leukotriene metabolite has to be followed if aspirin is used at all.

And the mirror caution: two of twenty patients in that aspirin series reacted during outpatient dose escalation, and a meaningful subset of mast cell patients are NSAID-reactive. An unqualified aspirin discussion is unsafe without that sentence. This is a physician-supervised intervention with a specific monitoring requirement, and I am describing it so you can have an informed conversation, not so you can start it.

**Omalizumab and KIT-directed therapy** are specialist-tier prescription treatments that appear in published algorithms (PMID 30961835). They are not something I prescribe, they are named here so you know the ladder does not end at supplements, and they are reached through allergy and immunology or hematology.

# 10. Safety: the reaction plan, the bleeding question, and when to escalate

---

## **The anaphylaxis plan**

If you have ever had a reaction with breathing difficulty, throat or tongue swelling, a drop in blood pressure, or fainting, the most important item in this entire program is an emergency action plan and an epinephrine autoinjector prescribed by a physician. That is standard of care. It is not a research question and it is not a functional medicine question.

Epinephrine autoinjectors are prescription devices. Prescribing them is not something I do, and they are named here so you know to ask for one from the physician managing that part of your care. If nobody has offered you one and you have had reactions like that, ask directly.

Around that prescription, the plan itself is education and I will help you build it: what your early signs are, what you do first, who you call, what you carry, what your household knows, and what happens after you use it, which is always emergency evaluation regardless of how much better you feel.

## **The bleeding question**

Per Chapter 2, mast cell granule heparin is real and is released on degranulation (PMID 26447996, PMID 39557214, PMID 12701115, PMID 34445665). Easy bruising and prolonged bleeding are mediator findings in this population, not incidental.

Two practical consequences. Tell any surgeon, dentist, or proceduralist about the mast cell picture in advance. And note that high-dose fish oil is mildly antiplatelet, so it is worth reviewing before a procedure. I want to be honest about the tiering here: the mastocytosis coagulopathy is documented, while clinically meaningful additive bleeding risk from typical supplement doses of fish oil is not. It is low probability and high consequence, which is exactly the kind of thing that costs nothing to mention and matters enormously the one time it applies.

## **When you leave me and go to a specialist**

There is a ladder, and I would rather you climb it than run another supplement round. Go to allergy and immunology or hematology when any of these is true: a persistently elevated baseline tryptase, especially with the alpha tryptasemia question unresolved; any documented mediator elevation on the urine panel; recurrent presyncope, syncope, or true anaphylaxis; unexplained low blood counts, bone pain, or an enlarged liver or spleen; or a picture that is escalating rather than holding.

I am saying this in my own book, about my own program, because pretending the ladder stops with me would be both a scope problem and a clinical one. Published algorithms include specialist-tier biologic and targeted

therapy for exactly these patients (PMID 30961835). If you need that tier, you need that tier, and I will help you get there rather than hold you.

## 11. The downstream branch: heavy periods, low iron, B12, and sleep

---

A large group of people in this space are exhausted, foggy, dizzy on standing, and losing hair, and have concluded it is all mediator burden. Some of it is. A great deal of it is iron, and the iron story has its own logic worth following, because it is more fixable than anything else in this book.

### What is actually established, and what is not

Mast cell activity in the endometrium is **normal**, not pathological. Across 107 specimens covering every day of the cycle, mast cells were present throughout and extensive degranulation was a common, ordinary feature just before and during menstruation (PMID 7492683). Any model treating that as inherently abnormal is starting from a false baseline.

Histamine acts on the uterus through two opposing receptors, and the predominant response of human myometrium is contraction (PMID 1358393, PMID 15547545). Uterine contraction is the main mechanism of menstrual hemostasis, so the muscular effect argues toward cramping rather than toward blood loss. The plausible bleeding route is vascular rather than muscular.

And here is where I will be straight with you: **histamine to heavy periods is the weakest link on this whole branch**. There is no trial, no cohort, not even a case series measuring menstrual blood loss in mast cell activation disease. The only quantitative human data linking endometrial mast cell counts to bleeding are two small studies from 1970 and 1974 in intrauterine device users, which is a foreign body inflammation model and not spontaneous mast cell activation (PMID 4193675). Biologically plausible, clinically unproven, and I would rather say that than sell you a mechanism.

### Where the link is strong

Heavy menstrual bleeding to iron deficiency is strong, well documented, and systematically under-recognized by patients and clinicians alike (PMID 36706856, PMID 40159291).

The reason this becomes a recurring problem rather than a one-time fix is arithmetic. Absorptive iron replacement tops out around 1 to 2 mg per day. As long as monthly loss exceeds that ceiling, repletion is temporary and relapse is guaranteed. Which means the answer is not only replacing iron, it is addressing the loss at the source, and that is a gynecological conversation with real, effective options.

### The route that is genuinely missing from most versions of this map

Menstrual blood loss does not meaningfully deplete B12. Stores run years and B12 is efficiently recycled. Putting B12 downstream of heavy periods is a physiology error.

The real route in this population is gastric, and it is iatrogenic. Long-term H2 receptor antagonists carry an odds ratio of 1.25 for B12 deficiency and proton pump inhibitors 1.65, dose-dependently (PMID 24327038). The same drugs carry odds ratios of 1.58 and 2.49 for iron deficiency, reaching 4.27 at high long-term PPI use (PMID 27890768).

Sit with that for a second. H1 and H2 blockade is the backbone of conventional mast cell management. The treatment for the upstream problem is itself a mechanism for both deficiencies downstream. That is not a reason to stop a medication that is working, and I am not telling you to stop anything. It is a reason to monitor B12 and iron in anyone on long-term acid suppression, which is not routinely done.

### **The other route to low iron, which does not involve losing any**

Inflammation drives hepcidin, which blocks iron absorption and traps iron in storage, producing iron-restricted red cell production **with a normal or high ferritin** (PMID 19786207). In a chronically activated patient, ferritin alone is unreliable. Transferrin saturation, soluble transferrin receptor, and a simultaneous CRP are what separate true depletion from sequestration.

### **The most important sentence in this chapter**

Fatigue responds to iron **without** anemia. A meta-analysis of six randomized controlled trials in iron deficiency without anemia found a pooled effect size of 0.33, with zero heterogeneity (PMID 28625177). The larger Cochrane review of 67 trials in 8,506 menstruating women found iron reduced anemia, raised hemoglobin, improved exercise performance and reduced symptomatic fatigue (PMID 27087396).

Which means hemoglobin is the wrong screening test. A normal hemoglobin does not clear you. Ferritin, transferrin saturation, and CRP do.

Downstream of that, the evidence tiers honestly. Cognitive change has support, with iron normalizing cognitive function in young women in one trial and Cochrane judging cognition inconsistent across trials (PMID 17344500). New-onset tinnitus shows a dose-response relationship with anemia severity in a matched cohort of women (PMID 41393945). Low iron stores and mild anemia are documented in postural tachycardia syndrome, which overlaps this population heavily (PMID 23720007). Cold hands, tension headache and muscle pain are weaker still and I will not present them as established.

Iron repletion itself is worth doing properly rather than aggressively. Riboflavin and vitamin A alongside iron reduced anemia prevalence in a randomized trial, which is a reminder that iron does not work alone (PMID 18806105). And timing matters: quercetin, polyphenols, calcium, zinc and binders all belong away from any iron dose.

### **Sleep apnea belongs upstream, not downstream**

Obstructive sleep apnea causes intermittent low oxygen, it is not caused by anemia, and the traffic runs the other way. In 527 patients with sleep apnea, over 98 percent had no elevated hemoglobin, because

inflammation-driven hepcidin limits iron availability and newly formed red cells are destroyed (PMID 37350302).

Untreated sleep apnea therefore contributes to iron-restricted red cell production and produces the exact fatigue, cognitive and headache picture being attributed to mediators. If you snore, wake unrefreshed, or wake with a headache, get screened. It is one of the highest-yield things on this list and it is not a supplement.

## 12. A ninety day roadmap

---

### **Days 1 to 14: gate and baseline, change almost nothing**

This phase feels slow and it is the phase that makes the rest interpretable.

Get the baselines drawn on a day you feel well: tryptase, CBC, comprehensive metabolic panel, ferritin with transferrin saturation, CRP, B12 with methylmalonic acid, 25-OH vitamin D, TSH. If it is available to you, the twenty-four hour urine mediator panel, collected as instructed.

Start the timed food and symptom diary from Chapter 13, because the popular biomarkers do not work and your own pattern is genuinely the better data.

Do the medication screen: any MAO inhibitor, any long-term acid suppression, anything on the MRGPRX2 list.

And answer the free question honestly: have antihistamines ever helped you, at a real dose, for a real trial length.

### **Days 15 to 30: control the input**

Now the diet levers from Chapter 6, run as a structured elimination with a defined end, not as a permanent way of eating. Storage and cold chain rules, which cost nothing. The alcohol decision. The explicit fermented food decision, made deliberately rather than by accident.

**Then hold it steady for two weeks before you change anything else.** This is the structural correction that matters most in this entire book. Introducing one agent every three to five days underneath an uncontrolled dietary input cannot read out. You will get false positives and false negatives, and every downstream decision will inherit them.

### **Days 31 to 60: raise capacity, one thing at a time**

Gut lining work. Copper at adequacy if a deficiency was found, with the zinc side reviewed. Vitamin D to target. The oral DAO trial if you are the right person for it, run as a trial with a stated endpoint and a stop date.

One agent at a time. Diluted, in a form you tolerate, introduced slowly.

### **Days 61 to 90: mediator-directed, by phenotype**

By now you know something you did not know on day one: which arm is driving. This is where the phenotype-directed work happens, inside the boundaries of Chapter 9, and where the prescription conversation happens with the physician who owns it.

And this is where reintroduction starts. Every restriction gets an end date. Breadth is a goal, not a failure.

### **Measure more than symptoms**

Symptoms are the point, and symptoms alone are a poor instrument in a condition with a 63 percent placebo response rate. Retest what you baselined. Track function: what you can eat, where you can go, what you can do, how you sleep, how many reaction days you had per month. Those are harder to fool.

# 13. Worksheets

---

## Worksheet 1: The one-page clinical summary

Bring this to every appointment. Fill it in once and update it.

- My top three symptoms, and what each one costs me
- Organ systems involved: skin, gut, airway, cardiovascular, neurological
- My baseline tryptase, and the date it was drawn on a well day
- Whether hereditary alpha tryptasemia has been assessed
- Whether antihistamines have ever helped, at what dose, for how long
- My urine mediator panel results, if obtained
- My ferritin, transferrin saturation, CRP, B12, methylmalonic acid, 25-OH vitamin D
- Current medications, including long-term acid suppression
- Reactions that required urgent care, with dates
- Whether I have an epinephrine autoinjector and an action plan

## Worksheet 2: The timed food and symptom diary

Four columns and nothing more, because a diary you actually keep beats a perfect one you abandon.

Time. What I ate or was exposed to. Symptom and severity from 0 to 10. Minutes from exposure to symptom.

Run it for at least fourteen days. The pattern you are looking for is timing, not blame: within minutes points one direction, three to six hours points another, and next-day points somewhere else entirely.

## Worksheet 3: The storage audit

- Do I buy fish fresh and eat it the same day
- Do leftovers get frozen rather than refrigerated
- How long does cooked protein sit in my refrigerator
- Do I know how long the fish counter has had that fish
- Are my highest-amine foods the ones sitting longest

## Worksheet 4: The reaction plan

- My earliest warning signs, in my own words

- What I do first
- When I use epinephrine, if prescribed
- Who I call, with the number
- What I carry, always
- Who in my household knows the plan and where it is written down

### **Worksheet 5: The single-agent trial log**

One row per agent. Start date. Dose and form. Diluted in how much water. What I expected to change. What actually changed by day 14. Kept or stopped, and why.

The rule that makes this worksheet work: one thing at a time, on a stable diet, for long enough to judge.

### **Worksheet 6: Reintroduction**

- Food, date reintroduced, portion, result
- Foods successfully returned to my diet
- Total number of foods I currently eat, tracked monthly, with the goal going **up**

That last row exists on purpose. In this condition, the number of foods you can eat is the outcome that matters most and the one most likely to quietly deteriorate.

## 14. Closing perspective

---

The thing I want you to take from this book is that the reason your last protocol did not work is probably not that you did it wrong, and it is probably not that you are an unusually severe case.

It is that this is at least two conditions wearing one set of symptoms. One of them is a clearance problem, driven by what arrives and how much enzyme you have. The other is an activation problem, driven by a cell that is releasing far more than histamine. They share a symptom list, they share an internet, and they do not share a treatment.

Sorting out which one you have, and which mediator is driving it, is a small number of specific decisions rather than an endless list. That work is doable, most of it is measurable, and it is what I would spend your first ninety days on.

And I would rather tell you that some of the arrows on the popular version of this map are unproven than hand you a tidy diagram that falls apart the first time it meets your labs.

# Working with Root Health

---

If you want help applying this, the next step is a conversation. The Root Discovery Session is free, and it exists to take your history, your labs, your medication list, and your actual symptom pattern and turn them into a specific plan rather than a general one. Book at [rootcauseunlocked.com/discovery](https://rootcauseunlocked.com/discovery).

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at [rootcauseunlocked.com/tracker](https://rootcauseunlocked.com/tracker) is free and works anywhere. In this condition specifically, tracking is not a nice extra. It is the instrument that tells you whether anything you are doing is working, in a situation where symptoms alone are a genuinely unreliable guide.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

**Where I source what I use.** I hold no physical inventory. I keep a screened dispensary through Fullscript, which ships directly to you, at [us.fullscript.com/welcome/drseay](https://us.fullscript.com/welcome/drseay). Root Health earns a portion of dispensary purchases. Anything I name in these pages that is not carried there is available from any reputable retailer, and I would rather you got the right product than the one with my link on it.

*This book is educational. It is not medical advice, diagnosis, or treatment, and reading it does not create a doctor and patient relationship.*

# References

---

1. Kanaoka Y, Urade Y. Hematopoietic prostaglandin D synthase. *Prostaglandins Leukot Essent Fatty Acids*. 2003. PMID 12895599.
2. Lyons JJ, et al. Elevated basal serum tryptase identifies a multisystem disorder associated with increased TPSAB1 copy number. *Nat Genet*. 2016. PMID 27749843.
3. Lyons JJ, et al. Genetically defined individual reference ranges for tryptase limit unnecessary procedures and unmask myeloid neoplasms. *Blood Adv*. 2023. PMID 36170795.
4. Greiner G, et al. Disease correlates and clinical relevance of hereditary alpha-tryptasemia in patients with systemic mastocytosis. *J Allergy Clin Immunol*. 2023. PMID 36309122.
5. Barete S, et al. A French National Survey on Clotting Disorders in Mastocytosis. *Medicine (Baltimore)*. 2015. PMID 26447996.
6. Thromboelastography for rapid diagnosis of heparin-like anticoagulant release during anaphylaxis-induced coagulopathy. *Clin Med (Lond)*. 2025. PMID 39557214.
7. Effects of human mast cell tryptase and eosinophil granule proteins on the kinetics of blood clotting. *Am J Hematol*. 2003. PMID 12701115.
8. Ravi A, et al. Nontryptase urinary and hematologic biomarkers of mast cell expansion and mast cell activation: status 2022. *J Allergy Clin Immunol Pract*. 2022. PMID 35346887.
9. Valent P, et al. Selecting the right criteria and proper classification to diagnose mast cell activation syndromes: a critical review. *J Allergy Clin Immunol Pract*. 2021. PMID 34166845.
10. AAAAI Mast Cell Disorders Committee Work Group Report: mast cell activation syndrome diagnosis and management. *J Allergy Clin Immunol*. 2019. PMID 31476322.
11. Weiler CR. Mast cell activation syndrome: tools for diagnosis and differential diagnosis. *J Allergy Clin Immunol Pract*. 2020. PMID 31470118.
12. Castells M, Butterfield J. Mast cell activation syndrome and mastocytosis: initial treatment options and long-term management. *J Allergy Clin Immunol Pract*. 2019. PMID 30961835.
13. Role of platelet-activating factor in the pathogenesis of chronic spontaneous urticaria. *Int J Mol Sci*. 2024. PMID 39596211.
14. Butterfield JH. Survey of aspirin administration in systemic mastocytosis. *Prostaglandins Other Lipid Mediat*. 2009. PMID 19429499.
15. Benefits and harms of aspirin desensitization for aspirin-exacerbated respiratory disease: a systematic review and meta-analysis. *Int Forum Allergy Rhinol*. 2019. PMID 31518069.
16. Divergent PGD2 and leukotriene C4 metabolite excretion following aspirin therapy: ten patients with systemic mastocytosis. *Prostaglandins Other Lipid Mediat*. 2021. PMID 34029712.
17. Siebenhaar F, et al. Rupatadine improves quality of life in mastocytosis: a randomized, double-blind, placebo-controlled trial. *Allergy*. 2013. PMID 23734572.
18. Efficacy and safety of up-dosing antihistamines in chronic spontaneous urticaria: a systematic review. *J Investig Allergol Clin Immunol*. 2021. PMID 33030434.
19. Frieri M, et al. Comparison of the therapeutic efficacy of cromolyn sodium with that of combined chlorpheniramine and cimetidine in systemic mastocytosis. *Am J Med*. 1985. PMID 3917606.

20. Horan RF, et al. Cromolyn sodium in the management of systemic mastocytosis. *J Allergy Clin Immunol*. 1990. PMID 2110198.
21. de Silva NL, et al. Leukotriene receptor antagonists for chronic urticaria: a systematic review. *Allergy Asthma Clin Immunol*. 2014. PMID 24817895.
22. Efficacy of leukotriene receptor antagonist with an anti-H1 receptor antagonist for treatment of chronic idiopathic urticaria. *J Dermatolog Treat*. 2009. PMID 19085267.
23. Briskey D, et al. The effect of Levagen+ (palmitoylethanolamide) supplementation on symptoms of allergic rhinitis: a double-blind placebo-controlled study. *Nutrients*. 2023. PMID 38068797.
24. McNeil BD, et al. Identification of a mast-cell-specific receptor crucial for pseudo-allergic drug reactions. *Nature*. 2015. PMID 25517090.
25. Boyden SE, et al. Vibratory urticaria associated with a missense variant in ADGRE2. *N Engl J Med*. 2016. PMID 26841242.
26. Stress and nasal allergy: corticotropin-releasing hormone stimulates mast cell degranulation and proliferation in human nasal mucosa. *Int J Mol Sci*. 2021. PMID 33803422.
27. Theoharides TC, et al. Corticotropin-releasing hormone induces skin mast cell degranulation and increased vascular permeability. *Endocrinology*. 1998. PMID 9421440.
28. Eggleston PA, et al. In vivo release of inflammatory mediators by hyperosmolar solutions. *Am Rev Respir Dis*. 1988. PMID 2449834.
29. Larsson J, et al. Mannitol triggers mast cell-dependent contractions of human small bronchi and prostacyclin bronchoprotection. *J Allergy Clin Immunol*. 2019. PMID 31207273.
30. Prostaglandin E2 prevents hyperosmolar-induced human mast cell activation through prostanoid receptors EP2 and EP4. *PLoS One*. 2014. PMID 25329458.
31. Liu ZQ, et al. Vitamin D contributes to mast cell stabilization. *Allergy*. 2017. PMID 27998003.
32. Wouters MM, et al. Impaired duodenal palmitoylethanolamide release underlies acid-induced mast cell activation in functional dyspepsia. *Cell Mol Gastroenterol Hepatol*. 2021. PMID 33065341.
33. Kabu K, et al. Zinc is required for FcepsilonRI-mediated mast cell activation. *J Immunol*. 2006. PMID 16818790.
34. Regulation of IgE-dependent zinc release from human mast cells. *Int Arch Allergy Immunol*. 2013. PMID 23711853.
35. Correlation of active site metal content in human diamine oxidase with trihydroxyphenylalanine quinone cofactor biogenesis. *Biochemistry*. 2010. PMID 20722416.
36. Human diamine oxidase cellular binding and internalization in vitro and rapid clearance in vivo. *Glycobiology*. 2021. PMID 32985651.
37. Wollin A, et al. Postheparin plasma diamine oxidase in health and intestinal disease. *Gastroenterology*. 1990. PMID 2110916.
38. Postheparin plasma diamine oxidase increases in patients with coeliac disease during gluten free diet. *Gut*. 1987. PMID 3121453.
39. Diamine oxidase in serum and small intestinal biopsy tissue in childhood celiac disease. *J Pediatr Gastroenterol Nutr*. 1986. PMID 3088248.
40. Maintz L, et al. Genetic variability of human diamine oxidase: occurrence of three nonsynonymous polymorphisms. *Pharmacogenet Genomics*. 2007. PMID 17700358.

41. Maintz L, et al. Association of single nucleotide polymorphisms in the diamine oxidase gene with diamine oxidase serum activities. *Allergy*. 2011. PMID 21488903.
42. Maintz L, et al. Effects of histamine and diamine oxidase activities on pregnancy: a critical review. *Hum Reprod Update*. 2008. PMID 18499706.
43. Response of diamine oxidase and other plasma copper biomarkers to various dietary copper intakes in the rat. *Br J Nutr*. 2000. PMID 10953681.
44. Characterization of recombinant human diamine oxidase produced in CHO cells. *J Biotechnol*. 2016. PMID 27063138.
45. Schnedl WJ, Enko D. Histamine intolerance originates in the gut. *Nutrients*. 2021. PMID 33921522.
46. Placebo-controlled histamine challenge disproves suspicion of histamine intolerance. *J Allergy Clin Immunol Pract*. 2023. PMID 37648152.
47. A popular myth: low-histamine diet improves chronic spontaneous urticaria, fact or fiction? *J Eur Acad Dermatol Venereol*. 2017. PMID 27624921.
48. Son JH, et al. A histamine-free diet is helpful for treatment of adult patients with chronic spontaneous urticaria. *Ann Dermatol*. 2018. PMID 29606813.
49. Measurement of diamine oxidase during low-histamine or ordinary diet in patients with histamine intolerance. *Eur J Clin Nutr*. 2024. PMID 38769188.
50. Yacoub MR, et al. Diamine oxidase supplementation in chronic spontaneous urticaria: a randomized, double-blind placebo-controlled study. *Int Arch Allergy Immunol*. 2018. PMID 29698966.
51. Izquierdo-Casas J, et al. Diamine oxidase supplement reduces headache in episodic migraine patients with DAO deficiency: a randomized double-blind trial. *Clin Nutr*. 2019. PMID 29475774.
52. Effect of diamine oxidase enzyme dietary supplementation in subjects with insomnia symptoms and AOC1 single nucleotide polymorphisms. *Sleep Med X*. 2026. PMID 41437955.
53. Diamine oxidase supplementation improves symptoms in patients with histamine intolerance. *Food Sci Biotechnol*. 2019. PMID 31807350.
54. McIntosh K, et al. FODMAPs alter symptoms and the metabolome of patients with IBS: a randomised controlled trial. *Gut*. 2017. PMID 26976734.
55. De Palma G, et al. Histamine production by the gut microbiota induces visceral hyperalgesia through histamine 4 receptor signaling in mice. *Sci Transl Med*. 2022. PMID 35895832.
56. Sanchez-Perez S, et al. Low-histamine diets: is the exclusion of foods justified by their histamine content? *Nutrients*. 2021. PMID 33919293.
57. The rate of histamine degradation by diamine oxidase is compromised by other biogenic amines. *Front Nutr*. 2022. PMID 35694170.
58. Diamine oxidase deficiency: implications for health, current management, and future directions. *Int J Biol Macromol*. 2025. PMID 40865824.
59. Emborg J, Dalgaard P. Significant histamine formation in tuna at 2 degrees C: effect of vacuum and modified atmosphere packaging. *Int J Food Microbiol*. 2005. PMID 15925710.
60. Growth, inactivation and histamine formation of *Morganella psychrotolerans* and *Morganella morganii*. *Int J Food Microbiol*. 2008. PMID 18845350.
61. Feng C, et al. Histamine (scombroid) fish poisoning: a comprehensive review. *Clin Rev Allergy Immunol*. 2016. PMID 25876709.

62. Wohrl S, et al. Histamine intolerance-like symptoms in healthy volunteers after oral provocation with liquid histamine. *Allergy Asthma Proc.* 2004. PMID 15603203.
63. Wastyk HC, et al. Gut-microbiota-targeted diets modulate human immune status. *Cell.* 2021. PMID 34256014.
64. Impact of the clinically approved *Petasites hybridus* extract Ze 339 on intestinal mechanisms involved in the handling of histamine. *Biomed Pharmacother.* 2022. PMID 35149385.
65. Schink M, et al. Microbial patterns in patients with histamine intolerance. *J Physiol Pharmacol.* 2018. PMID 30552302.
66. Sanchez-Perez S, et al. Intestinal dysbiosis in patients with histamine intolerance. *Nutrients.* 2022. PMID 35565742.
67. Diamine oxidase interactions with anti-inflammatory and anti-migraine medicines in the treatment of migraine. *J Clin Med.* 2023. PMID 38068554.
68. Recombinant human diamine oxidase activity is not inhibited by ethanol, acetaldehyde, disulfiram, diethyldithiocarbamate. *Alcohol.* 2016. PMID 27401969.
69. Food-induced histaminosis as an epidemiological problem. *Agents Actions.* 1988. PMID 3134804.
70. Food-induced histaminosis under diamine oxidase blockade in pigs. *Agents Actions.* 1989. PMID 2568741.
71. Intestinal diamine oxidases and enteral-induced histaminosis. *J Neural Transm Suppl.* 1990. PMID 2128501.
72. Azzaro AJ, et al. Effects of a tyramine-enriched meal on blood pressure response in healthy male volunteers treated with selegiline transdermal system. *CNS Spectr.* 2007. PMID 17192761.
73. McCabe BJ. Dietary tyramine and other pressor amines in MAOI regimens: a review. *J Am Diet Assoc.* 1986. PMID 3525654.
74. Yamauchi K, et al. Structure and function of human histamine N-methyltransferase: critical enzyme in histamine metabolism in airway. *Am J Physiol.* 1994. PMID 7943261.
75. Histamine clearance through polyspecific transporters in the brain. *Handb Exp Pharmacol.* 2017. PMID 27679412.
76. Organic cation transporters in brain histamine clearance. *Handb Exp Pharmacol.* 2021. PMID 33641029.
77. Francis DM, et al. The kinetic properties and reaction mechanism of histamine methyltransferase from human skin. *Biochem J.* 1980. PMID 7188427.
78. Preuss CV, et al. Human histamine N-methyltransferase pharmacogenetics: common genetic polymorphisms that alter activity. *Mol Pharmacol.* 1998. PMID 9547362.
79. Horton JR, et al. Two polymorphic forms of human histamine methyltransferase: structural, thermal, and kinetic comparisons. *Structure.* 2001. PMID 11566133.
80. Heidari A, et al. Mutations in the histamine N-methyltransferase gene, HNMT, are associated with nonsyndromic autosomal recessive intellectual disability. *Hum Mol Genet.* 2015. PMID 26206890.
81. Impact of the MTHFR C677T polymorphism on one-carbon metabolites: evidence from a randomised trial of riboflavin supplementation. *Biochimie.* 2020. PMID 32330571.
82. Goulding CW, Matthews RG. Cobalamin-dependent methionine synthase: involvement of zinc in homocysteine activation. *Biochemistry.* 1997. PMID 9398304.
83. Millian NS, Garrow TA. Human betaine-homocysteine methyltransferase is a zinc metalloenzyme. *Arch Biochem Biophys.* 1998. PMID 9681996.
84. Yeo EJ, Wagner C. Structure, function and physiological role of glycine N-methyltransferase. *Int J Biochem Cell Biol.* 1998. PMID 9597750.

85. Regulation of folate-mediated one-carbon metabolism by glycine N-methyltransferase and MTHFR. *J Nutr Sci Vitaminol (Tokyo)*. 2015. PMID 26598833.
86. Pissios P. Nicotinamide N-methyltransferase: more than a vitamin B3 clearance enzyme. *Trends Endocrinol Metab*. 2017. PMID 28291578.
87. Kraus D, et al. Nicotinamide N-methyltransferase knockdown protects against diet-induced obesity. *Nature*. 2014. PMID 24717514.
88. Lowering plasma S-adenosylhomocysteine in healthy adults with elevated SAH and normal homocysteine using nutritional supplementation. *Nutr Metab Cardiovasc Dis*. 2025. PMID 40883125.
89. Comparison between number of basophils, blood histamine, and histamine release. *J Allergy Clin Immunol*. 1989. PMID 2477430.
90. Value of urinary N-methylhistamine measurements in childhood mastocytosis. *J Am Acad Dermatol*. 1996. PMID 8859283.
91. Biomarkers in the diagnosis of mast cell activation. *Curr Opin Allergy Clin Immunol*. 2025. PMID 39745330.
92. Jeziorska M, et al. Mast cell and eosinophil distribution and activation in human endometrium throughout the menstrual cycle. *Biol Reprod*. 1995. PMID 7492683.
93. Effect of histamine and histamine analogues on human isolated myometrial strips. *Br J Pharmacol*. 1992. PMID 1358393.
94. Degranulation of uterine mast cell modifies contractility of isolated myometrium from pregnant women. *Am J Obstet Gynecol*. 2004. PMID 15547545.
95. Effects of primary mast cell disease on hemostasis and erythropoiesis. *Int J Mol Sci*. 2021. PMID 34445665.
96. Sheppard BL, Bonnar J. The relationship between endometrial mast cell count and bleeding in women following insertion of an intrauterine device. *Am J Obstet Gynecol*. 1970. PMID 4193675.
97. The relationship between heavy menstrual bleeding, iron deficiency, and iron deficiency anemia. *Am J Obstet Gynecol*. 2023. PMID 36706856.
98. Iron deficiency in adults: a review. *JAMA*. 2025. PMID 40159291.
99. Lam JR, et al. Proton pump inhibitor and histamine 2 receptor antagonist use and vitamin B12 deficiency. *JAMA*. 2013. PMID 24327038.
100. Tran-Duy A, et al. Proton pump inhibitor and histamine-2 receptor antagonist use and iron deficiency. *Gastroenterology*. 2017. PMID 27890768.
101. Ganz T, Nemeth E. Iron sequestration and anemia of inflammation. *Semin Hematol*. 2009. PMID 19786207.
102. Low MSY, et al. Daily iron supplementation for improving anaemia, iron status and health in menstruating women. *Cochrane Database Syst Rev*. 2016. PMID 27087396.
103. Houston BL, et al. Iron deficiency without anaemia is a potential cause of fatigue: meta-analyses of randomised controlled trials. *Br J Nutr*. 2017. PMID 28625177.
104. Murray-Kolb LE, Beard JL. Iron treatment normalizes cognitive functioning in young women. *Am J Clin Nutr*. 2007. PMID 17344500.
105. Iron deficiency anemia and the risk of new-onset tinnitus in female patients: a cohort study. *Front Nutr*. 2025. PMID 41393945.
106. Jarjour IT, Jarjour LK. Low iron storage and mild anemia in postural tachycardia syndrome in adolescents. *Clin Auton Res*. 2013. PMID 23720007.

107. Increased blood reactive oxygen species and hepcidin in obstructive sleep apnea precludes expected erythrocytosis. *Am J Hematol*. 2023. PMID 37350302.
108. Prevalence of heavy menstrual bleeding, iron deficiency, iron deficiency anemia, and treatment in women with von Willebrand disease. *Res Pract Thromb Haemost*. 2025. PMID 40697484.
109. Prevalence and impact of heavy menstrual bleeding in women with von Willebrand disease across age groups. *Haemophilia*. 2025. PMID 40955497.
110. Hershko C, et al. Role of autoimmune gastritis, *Helicobacter pylori* and celiac disease in refractory or unexplained iron deficiency anemia. *Haematologica*. 2005. PMID 15921373.
111. Camaschella C. How I treat unexplained refractory iron deficiency anemia. *Blood*. 2014. PMID 24215034.
112. Double-blind placebo-controlled provocation study in patients with subjective multiple chemical sensitivity. *Clin Toxicol (Phila)*. 2008. PMID 18568800.
113. A novel herbal composition containing extracts of *Boswellia serrata* gum resin and *Aegle marmelos* fruit alleviates symptoms of asthma. *Phytother Res*. 2018. PMID 29210124.
114. Schapowal A. Randomised controlled trial of butterbur and cetirizine for treating seasonal allergic rhinitis. *BMJ*. 2002. PMID 11799030.
115. Schapowal A. Butterbur Ze339 for the treatment of intermittent allergic rhinitis: dose-dependent efficacy. *Arch Otolaryngol Head Neck Surg*. 2004. PMID 15611396.
116. Jackson CM, et al. The effects of butterbur on the histamine and allergen cutaneous response. *Ann Allergy Asthma Immunol*. 2004. PMID 14989395.
117. Rasool R, et al. Chronic urticaria merits serum vitamin D evaluation and supplementation: a randomized case control study. *World Allergy Organ J*. 2015. PMID 26097642.
118. Brannan JD, et al. The effect of omega-3 fatty acids on bronchial hyperresponsiveness, sputum eosinophilia, and mast cell mediators in asthma. *Chest*. 2015. PMID 25321659.
119. Kawai M, et al. Effect of enzymatically modified isoquercitrin on symptoms of Japanese cedar pollinosis: a randomized double-blind placebo-controlled trial. *Int Arch Allergy Immunol*. 2009. PMID 19295240.
120. Effects of repeated oral intake of a quercetin-containing supplement on allergic reaction: a randomized, placebo-controlled trial. *Eur Rev Med Pharmacol Sci*. 2022. PMID 35776034.
121. Pyridoxal-5'-phosphate as a cofactor for rat brain histidine decarboxylase. *J Neurochem*. 1978. PMID 413882.
122. Properties of histidine decarboxylase from rat gastric mucosa. *Eur J Biochem*. 1982. PMID 7075603.
123. Maintz L, Novak N. Histamine and histamine intolerance. *Am J Clin Nutr*. 2007. PMID 17490952.
124. Landete JM, et al. Updated molecular knowledge about histamine biosynthesis by bacteria. *Crit Rev Food Sci Nutr*. 2008. PMID 18756395.
125. Pyruvoyl-dependent histidine decarboxylases: preparation and amino acid sequences. *J Biol Chem*. 1985. PMID 2857718.
126. Pyruvoyl-dependent histidine decarboxylase from *Lactobacillus 30a*: covalent modifications. *J Biol Chem*. 1986. PMID 3082865.
127. Wessler S, et al. *Helicobacter pylori* activates the histidine decarboxylase promoter through a MAP kinase pathway. *J Biol Chem*. 2000. PMID 10652359.
128. Histamine content of the oxyntic mucosa from duodenal ulcer patients: effect of *Helicobacter pylori* eradication. *Am J Gastroenterol*. 1993. PMID 8338090.

- 129.** *Helicobacter pylori* and gastric histamine concentrations. *J Clin Pathol.* 1991. PMID 1856297.
- 130.** Retinol and riboflavin supplementation decreases the prevalence of anemia in Chinese pregnant women taking iron and folic acid. *J Nutr.* 2008. PMID 18806105.
- 131.** Das-Munshi J, Rubin GJ, Wessely S. Multiple chemical sensitivities: a systematic review of provocation studies. *J Allergy Clin Immunol.* 2006. PMID 17137865.
- 132.** Kimata H. Effect of exposure to volatile organic compounds on plasma levels of neuropeptides, nerve growth factor and histamine in patients with self-reported multiple chemical sensitivity. *Int J Hyg Environ Health.* 2004. PMID 15031958.
- 133.** Ternesten-Hasseus E, Bende M, Millqvist E. Increased capsaicin cough sensitivity in patients with multiple chemical sensitivity. *J Occup Environ Med.* 2002. PMID 12448352.
- 134.** Tran MT, et al. Multiple chemical sensitivity: on the scent of central sensitization. *Int J Hyg Environ Health.* 2013. PMID 22487274.